

1. Administrative & Study Identification

Full Study Title: Assessment of 13-valent Pneumococcal Conjugate Vaccine Effectiveness Among People With HIV in the United States
Short Title/Acronym: A Study To Learn About The Effects Of Pneumococcal Vaccine In People With HIV
Protocol Number: B1851217
NCT Number: NCT05794191
Sponsor Name: Pfizer
Investigational Product: 13-valent Pneumococcal Conjugate Vaccine (Prevnar 13, PCV13)
Phase of Development: Observational / N/A for Secondary Data Collection
Medical Monitor: Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective
Primary Objective: To estimate 13-valent pneumococcal conjugate vaccine (PCV13) vaccine effectiveness (VE) against invasive pneumococcal disease (IPD) in people living with HIV (PLWH) aged ≥ 18 years, overall and at 3, 5, and 7 years of follow-up.
Secondary Objectives: To estimate the VE of PCV13 for pneumococcal pneumonia and all-cause pneumonia among PLWH ≥ 18 years of age.

2.2 Background & Rationale People living with HIV (PLWH) are at an increased risk of pneumococcal infections, but there is a lack of data on PCV13 vaccine effectiveness (VE) in this specific population. This study utilizes administrative claims and healthcare databases to understand how well the vaccine works in preventing pneumonia—especially pneumococcus-caused pneumonia and bloodstream infections—in adult PLWH in a real-world setting.

3. Study Design & Methodology

3.1 Design Framework
Study Type: Observational (Retrospective secondary data collection study)
Control Type: Unvaccinated PLWH (PCV13 vaccination status evaluated as a time-varying exposure)
Blinding: Open-label / N/A (Database study)
Randomization: N/A (Propensity scoring and Inverse Probability of Treatment Weighting [IPTW] used to reduce confounder imbalances)

3.2 Planned Enrollment & Duration
Target N: Approximately 175,000 individuals meeting eligibility criteria in the Komodo Health database
Number of Sites: 1 (Pfizer Investigational Site / Centralized Database Analysis)
Estimated Study Start: 22-Mar-2023 (Actual)
Estimated Primary Completion: 15-Aug-2024 (Actual)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. HIV infection defined as at least one inpatient ICD-9-CM or ICD-10-CM code related to HIV, or ≥ 2 outpatient codes at least 30 days but no more than 730 days apart during January 1, 2014, and December 31, 2021. 2. At least 18 years of age at the time of the first recording of an HIV-related ICD-9-CM or ICD-10-CM code. 3. At least six months of continuous enrollment in medical and pharmacy plans after the first HIV-related code (a gap of ≤ 30 days allowed).

4.2 Exclusion Criteria 1. Evidence of PCV13 vaccination before the first HIV-related ICD-9-CM or ICD-10-CM code. 2. Inadequate continuous enrollment in healthcare/pharmacy plans during the required baseline period.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single administration of the 13-valent pneumococcal conjugate vaccine (PCV13) as recorded in medical/pharmacy claims. **Route of Administration:** Intramuscular Injection (Vaccine Administration). **Duration of Treatment:** Single administration with long-term observational follow-up up to 7 years.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care for HIV management, including antiretroviral therapy (ART) and other routine medical treatments identified in the database. **Prohibited:** Receipt of PCV13 prior to the first diagnostic code for HIV.

6. Study Timeline & Visit Schedule

(Note: As a secondary database study, timeline reflects data collection windows rather than physical visits) | Visit ID | Window (Days) | Key Activities/Procedures | | :--- | :--- | :---
| | **Screening** | Jan 2014 to Dec 2021 | Identification of HIV-related diagnostic codes and PCV13 vaccination status in the Komodo Health database | | **Baseline** | First 6 months | Continuous enrollment period to assess baseline covariates, demographics, and comorbidities | | **Interim** | Years 3 and 5 | Database assessment for incidence of invasive pneumococcal disease and pneumonia | | **End of Study** | Year 7 / End of Data | Final data extraction (up to Sept 30, 2022) to calculate long-term vaccine effectiveness |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Vaccine Effectiveness (VE) against Invasive Pneumococcal Disease (IPD), calculated as $1 -$

$\text{Hazard Ratio} \times 100\%$. **Measurement Method:** Identified using ICD-9-CM or ICD-10-CM diagnosis codes from administrative healthcare claims, analyzed via Cox Marginal Structural Models (Cox-MSM).

7.2 Secondary & Exploratory Endpoints 1. Vaccine effectiveness (VE) of PCV13 for pneumococcal pneumonia. 2. Vaccine effectiveness (VE) of PCV13 for all-cause pneumonia.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: N/A (This is a retrospective non-interventional claims database study evaluating disease incidence rather than prospectively collecting direct safety/AE reports). **Serious Adverse Events (SAEs):** Evaluated strictly within the context of the study's primary outcomes (e.g., hospitalization for invasive pneumococcal disease). **Data Monitoring Committee (DMC):** Internal non-interventional study review by Pfizer.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All eligible PLWH identified in the Komodo Health administrative claims dataset (~175,000 individuals). **Sample Size Justification:** Based on the availability of PLWH meeting inclusion criteria in the target database; powered by the large cohort size for robust longitudinal analysis. **Handling of Missing Data:** Inverse probability of censoring weighting (IPCW) was used to adjust for informative censoring. Inverse probability of treatment weighting (IPTW) derived from propensity scores was used to create pseudo-populations, reducing imbalances in potential confounders by vaccination status.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study is conducted in accordance with Good Pharmacoepidemiology Practices (GPP) and applicable regulatory requirements for non-interventional secondary data studies. **Monitoring Plan:** Algorithmic validation of ICD-9-CM, ICD-10-CM, and CPT codes (e.g., assessing annual HIV viral load testing and CD4 cell count testing) within the administrative claims database. **Audit Trail:** Detailed tracking of patient eligibility attrition and data extraction queries per the Pfizer Statistical Analysis Plan (CT24-WI-GL03-RF03).

11. Study Identifiers & Governance

Primary ID: B1851217 **Registry Number:** NCT05794191 **Sponsor/Lead Organization:** Pfizer **Study Title:** A Study To Learn About The

12. Clinical Framework (Condition Specific)

Primary Condition: HIV, Pneumonia, and Pneumococcal Infections
Diagnosis Threshold: ≥ 1 inpatient or ≥ 2 outpatient
ICD-9/10 codes for HIV within a 30 to 730-day window **Medical**
Methodology: Retrospective health care administrative claims and
laboratory data analysis **Optimization Target:** Real-world PCV13
vaccine effectiveness (VE) over 3, 5, and 7 years

13. Study Procedures & Methodology

Management Pathway: Standard clinical care for PLWH evaluated
retrospectively **Education Protocol (HCP):** N/A (Retrospective
Database Study) **Education Protocol (Patient):** N/A (Retrospective
Database Study) **Quality Audit Mechanism:** Verification of
continuous medical and pharmacy plan enrollment (minimum of 6
months post-HIV diagnosis) to ensure data completeness.

14. Observation & Follow-Up Schedule

Total Duration: Up to 7 years of observational follow-up **Onsite**
Visit Cadence: N/A (Data extracted from claims database) **Remote**
Monitoring Frequency: Continuous longitudinal database tracking
Checklist Review Interval: Data evaluated at 3, 5, and 7-year
follow-up milestones for incident disease

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	Pfizer Study Director	_____	02-				
Apr-2026 	Clinical Operations Lead 	Pfizer Operations Lead 					
_____	02-Apr-2026 	Lead Statistician 	Pfizer Lead				
Biostatistician 	_____	02-Apr-2026 					